agents such as cetrimide, bromide and benzalkonium chloride. Their antibacterial activity is antagonized by soaps and certain organisms like *Pseudomonas*. They are useful for washing cutlery in catering industry and for cleaning wounds in hospitals. Savlon, a popular antiseptic, is a mixture of cetrimide and chlorohexidine and is active against Gram negative bacteria. They are used as skin disinfectants and as a preservative of ophthalmic solution.

The combined properties of germicidal activity and low toxicity, high solubility in water, stability in solution and non-corrosiveness have resulted in many applications of quaterneries as disinfectants and sanitizing agents.

Quats are also fungicidal, amoebicidal, and virucidal against enveloped viruses. They do not kill endospores or mycobacteria.

Infobits

If you mouthwash bottle fills with foam when shaken, the mouthwash probably contains a quat.

6. Aldehydes

Aldehydes are highly effective, broad spectrum disinfectant. The most which typically achieve its anitimicrobial action by denaturing proteins and disrupting nucleic acids. Commonly used aldehydes are formaldehyde and glutaraldehyde. Formaldehyde is usually dissolved in water or alcohol before use for moldehyde is used as a surface disinfectat and a fumigant and has been used to decontaminate in animate objects. A concentration 2% of glutaraldehyde is an effective disinfectant. It is less irritating than formaldehyde and is used to disinfect hospital and laboratory equipments. Glutaraldehyde usually disinfects objects within about 10 minutes but may require as long as 12 hours to destroy all spores.

Infobits

Disinfection of Rooms

Fumigation with gaseous disinfectants was at one time commonly performed after a room had been occupied by a patient with an infections disease. Sulphur di oxide, generated by burning sulphur was the popular agent for this purpose but it is effective only if the relative humidity is 60 percent or more.

These are highly reactive molecules that combine with nucleic acids and proteins and inactivate them. They disrupt the function of cell organelles and kill the cells probably by cross linking and alkylating the molecules. These are sporicidal and can be used as chemical sterilants.

7. Gaseous Sterilization

Gaseous disinfectants (alkylating agents) are used for the sterilization or disinfection of hospital equipment that is bulky or heat labile. The most widely used gases are ethylene oxide, formaldehyde and β Propiolactone.

Ethylene oxide (EtO)

Ethylene oxide has a boiling point of 10.8°C. It is highly inflammable and explosive in pure form, but is safe to handle when mixed with Carbon dioxide. It is powerful in the killing of all bacteria,

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